# Architecture Selection: Micro-Lending & Peer Credit Risk Assessor

Lab 3: Component Modelling and Architectural Pattern Selection | PES2UG24CS098
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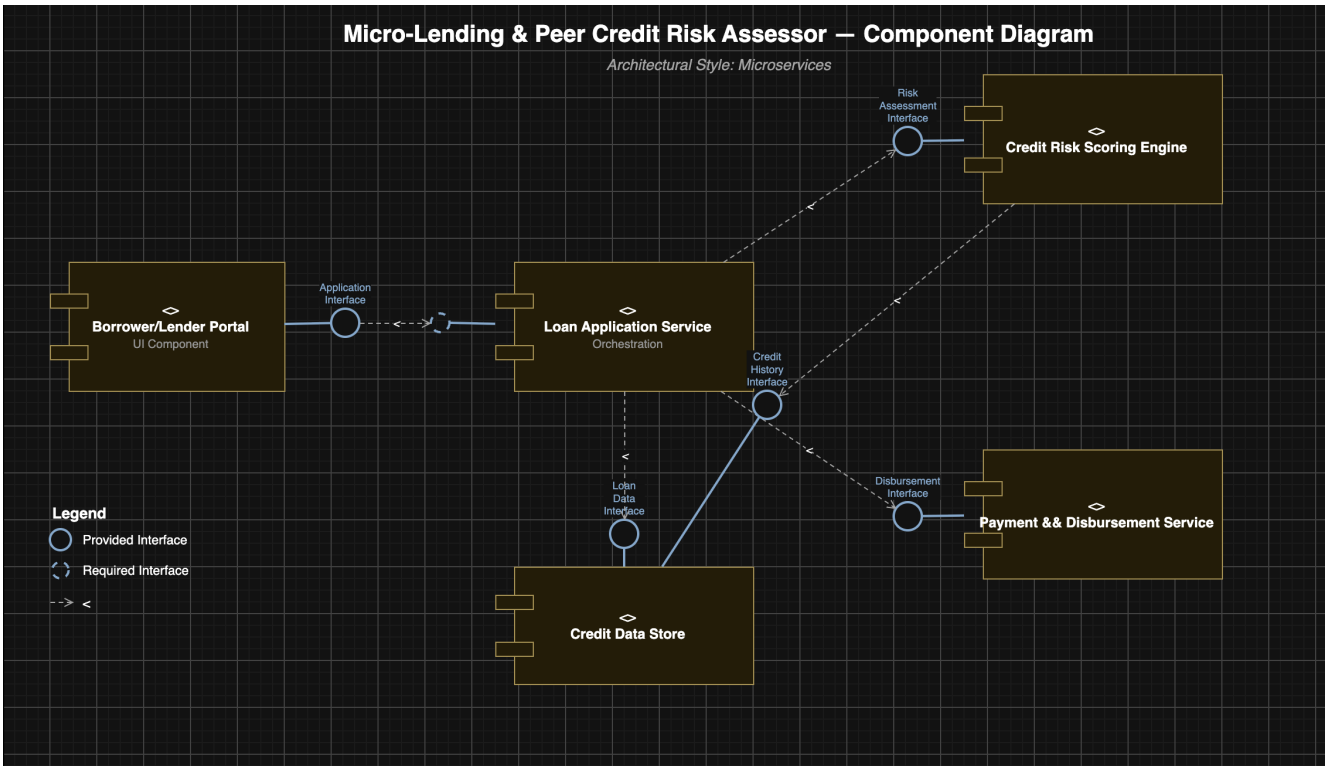



*Figure 1: UML component diagram (microservices architecture) for the Micro-Lending & Peer Credit Risk Assessor.*

We chose Microservices Architecture for the Micro-Lending & Peer Credit Risk Assessor. The system is split into five independently deployable components: the Borrower/Lender Portal, the Loan Application Service, the Credit Risk Scoring Engine, the Credit Data Store and the Payment & Disbursement Service.

## Reason 1: Two workloads with opposite shapes

NFR-001 requires marketplace listings, remaining amounts and risk scores to update in real time with an API latency under 300 ms. This is a read heavy and write heavy path that lenders hit constantly while they browse and pledge funds (FR-003). Credit risk scoring (FR-002) and fraud screening (NFR-002) are the opposite: they pull in repayment history, bank cash flow patterns and peer endorsements, run heavier computation, and only fire when a request is submitted. Microservices let us run many replicas of the Loan Application Service for the marketplace, while the Credit Risk Scoring Engine runs on its own compute optimised nodes. The scoring result is stored with the listing, so lenders browsing the marketplace never trigger a score computation. A layered or client server design ships both in one deployable unit and cannot scale one side without the other.

## Reason 2: Independent change and failure isolation

The scoring model is the part of the system most likely to change, because the Credit Risk Analyst will keep tuning thresholds and data sources to improve grades and fraud detection. As a separate component it can be redeployed without touching the marketplace. Failures are also contained. Disbursement (FR-004) depends on an external bank or payment gateway, so an outage there delays payouts only, while borrowers can still submit requests and lenders can still pledge. In the same way, a fault in the Credit Risk Scoring Engine stops new listings from being published (FR-002) but does not stop existing listings from being funded or repayments from being recorded (FR-005).

## Security Advantage

The Borrower/Lender Portal is the only component with a public route. It terminates TLS and forwards every call to the Loan Application Service, which authenticates the caller and checks their role before doing anything. The Credit Risk Scoring Engine, Credit Data Store and Payment & Disbursement Service sit on a private network with no inbound route from the internet, and the services prove their identity to each other with mutual TLS, so a stolen session token is rejected at the service boundary rather than at the database. Each service holds its own database account scoped to its own data, so the Loan Application Service can record pledges but cannot alter the stored risk grade or the audit log of contributing factors required by FR-002, and the scoring engine cannot move money. The 20% concentration cap (FR-003) is enforced on the server inside the Loan Application Service, never in the portal. Funds are released only by the Payment & Disbursement Service, and only for a fully funded loan whose OTP was verified within the 5 minute window (FR-004). Sensitive KYC and bank linkage data stays in the Credit Data Store behind that private network. The result is a small blast radius: compromising the public portal does not give an attacker the ability to bypass KYC checks (FR-001), breach the exposure cap, or trigger a disbursement.

## Performance Benefit

The risk grade is computed once, through the Risk Assessment Interface, when a loan request is submitted, and is then stored and attached to the listing. Lender requests for listings and remaining amounts go through the Loan Data Interface and are served by replicas of the Loan Application Service backed by a cache, which keeps the marketplace within the 300 ms target in NFR-001 even while many requests are being scored at the same time. Replicas can be added for the marketplace alone, without resizing the Credit Data Store or the scoring nodes. Repayment recording and status updates (FR-005) run against the same data store and do not compete with the scoring workload for compute.

## Disadvantages Accepted

1. Infrastructure cost and management effort. For a platform of this size, the container platform, service registry, gateway, message broker and per service monitoring cost more in money and staff time than one deployable application would. We accept this overhead because the marketplace and the scoring engine genuinely need to scale and change separately.
2. Consistency is harder to hold. A pledge touches the remaining amount, the 20% cap check and, once the loan is fully funded, the disbursement step in the Payment & Disbursement Service. These span more than one service, so the design needs idempotent handlers and compensating actions in place of a single database transaction. There are also brief windows where a lender could see a remaining amount that is slightly stale, so the pledge check must always be made against the Credit Data Store rather than against the cached value.